

Problem A. Range Update/Point Query

Input file: **standard input**
Output file: **standard output**
Time limit: 15 seconds
Memory limit: 256 megabytes

Given a sequence $a[1], a[2], \dots, a[n]$, process the following q operations:

- 1 l r x: Given l, r, x , add x to all $a[i], i \in [l, r]$.
- 2 i: Output $a[i]$.

Input

The first line contains $n, q (1 \leq n, q \leq 10^6)$.

The second line contains $a[1], a[2], \dots, a[n]$, guaranteed $|a[i]| \leq 10^6$.

The following q lines contain one of the two types of operations, guaranteed $1 \leq l \leq r \leq n, |x| \leq 10^6$.

Output

Output the answer for each operation 2.

Example

standard input	standard output
3 2 1 2 3 1 1 3 0 2 2	2

Problem B. Range Update/Range Query

Input file: **standard input**
Output file: **standard output**
Time limit: 15 seconds
Memory limit: 1024 megabytes

Given a sequence $a[1], a[2], \dots, a[n]$, process the following q operations:

- 1 l r x: Given l, r, x , add x to all $a[i], i \in [l, r]$.
- 2 l r: Output $\sum_{i=l}^r a[i]$.

Input

The first line contains $n, q (1 \leq n, q \leq 10^6)$.

The second line contains $a[1], a[2], \dots, a[n]$, guaranteed $|a[i]| \leq 10^6$.

The following q lines contain one of the two types of operations, guaranteed $1 \leq l \leq r \leq n, |x| \leq 10^6$.

Output

Output the answer for each operation 2.

Example

standard input	standard output
5 10	15
2 6 6 1 1	34
2 1 4	32
1 2 5 10	33
2 1 3	50
2 2 3	
1 2 2 8	
1 2 3 7	
1 4 4 10	
2 1 2	
1 4 5 6	
2 3 4	

Problem C. Range Maximum Subarray

Input file: **standard input**
Output file: **standard output**
Time limit: 5 seconds
Memory limit: 256 megabytes

You are given a sequence $A[1], A[2], \dots, A[N]$. ($|A[i]| \leq 15007, 1 \leq N \leq 50000$). A query is defined as follows: $\text{Query}(x,y) = \text{Max}\{a[i] + a[i + 1] + \dots + a[j]; x \leq i \leq j \leq y\}$. Given M queries, your program must output the results of these queries.

Input

The first line of the input file contains the integer N . In the second line, N numbers follow. The third line contains the integer $M \leq 50000$. M lines follow, where line i contains 2 numbers x_i and y_i .

Output

Your program should output the results of the M queries, one query per line.

Example

standard input	standard output
3 -1 2 3 1 1 2	2

Problem D. Rectangle Area Union

Input file: **standard input**
Output file: **standard output**
Time limit: 10 seconds
Memory limit: 256 megabytes

Given n rectangle that is parallel to axis, compute the union area.

Input

The first line contains an integer n ($1 \leq n \leq 10^5$), indicating the number of rectangles. The following n lines contains x_1, y_1, x_2, y_2 ($0 \leq x_1 < x_2 \leq 10^9, 0 \leq y_1 < y_2 \leq 10^9$).

Output

Output the area union.

Example

standard input	standard output
2 100 100 200 200 150 150 250 255	18000

Note

10^9 is too big for building a $[1, 10^9]$ segment tree, you need to shrink it to $[1, n]$.